SoS 2026: Stochastic Processes
Plan of Action

Application Track: Markov Chain Monte Carlo

Mentee: Dhruv Natu

May 2026

Timeline

Week 1 Foundations of probability: events, axioms of probability, conditional probability, independence, and Bayes' rule. Discrete random variables, probability mass functions, expectation, and linearity of expectation. Working through introductory examples including the coupon collector's problem and analysis of randomized algorithms to build comfort with probabilistic reasoning. *Primary readings: Mitzenmacher & Upfal, Chapters 1 and 2.*

Week 2 Moments and deviations: variance, higher moments, Markov's inequality, Chebyshev's inequality, and a first taste of concentration of measure. Brief introduction to Chernoff-type bounds. By the end of the week, a first informal introduction to stochastic processes and the notion of a Markov chain — "getting a vibe" for what Markov chains are before the formal treatment in Week 3. *Primary readings: Mitzenmacher & Upfal, Chapter 3 (and §4.1–4.2 for a glimpse of Chernoff); Norris §1.1 for the informal Markov chain setup.*

Week 3 Discrete-time Markov chains, formal treatment: state spaces, transition matrices, n-step transition probabilities, and the Chapman–Kolmogorov equations. The strong Markov property and first-step analysis. Numerous worked examples on small state spaces to build intuition before the more delicate convergence theory in Weeks 4–5. *Primary readings: Norris, Chapter 1 (§1.1–1.4); Mitzenmacher & Upfal §7.1–7.2.*

Week 4 Classification of states and long-run behaviour: communicating classes, irreducibility, recurrence vs. transience, positive vs. null recurrence, and periodicity. Hitting times and absorption probabilities. The relationship between detailed balance and reversibility, and why reversibility is a natural sufficient condition for stationarity. *Primary readings: Norris, Chapter 1 (§1.5–1.9); Mitzenmacher & Upfal §7.3.*

Mid-term Report Submission

Week 5 Convergence of Markov chains and the ergodic theorem: existence and uniqueness of stationary distributions, convergence to equilibrium for irreducible aperiodic chains, and the ergodic theorem (statement, intuition, and proof sketch). Mixing times and a brief look at how convergence rates are quantified in practice. *Primary readings: Norris, Chapter 1 (§1.10) and Chapter 3; Mitzenmacher & Upfal §7.4–7.5.*

Week 6 Introduction to martingales: filtrations, conditional expectation in the abstract sense, and the definition of a discrete-time martingale (and the related notions of submartingale and supermartingale). Foundational examples including random walks, the Doob martingale, and martingales arising from sums of independent zero-mean random variables. Building comfort with the definitions through worked examples before the powerful theorems of Week 7. *Primary readings: Mitzenmacher & Upfal, Chapter 12 (§12.1–12.2); Williams, Probability with*

Martingales, Chapters 9–10.

Week 7 Stopping times and the optional stopping theorem: the formal definition of a stopping time, statement of Doob's optional stopping theorem with its hypotheses, and classical applications including the gambler's ruin problem and Wald's identity. Concentration inequalities from martingales: the Azuma–Hoeffding inequality and applications to the analysis of randomized algorithms. *Primary readings: Mitzenmacher & Upfal, Chapter 12 (§12.3–12.5); Williams, Probability with Martingales, Chapters 10–11.*

Week 8 Wrap-up and flexible week. Synthesis of the key results from the three pillars of the project (probability foundations, Markov chains, martingales). Depending on interest, this week may also be used to revisit any earlier topic that needs strengthening, explore a short application chapter, or work through additional problems in the quantitative-puzzle direction. Report writing and video presentation preparation occur in parallel.

End-term Report Submission (July 27)

References

1. Michael Mitzenmacher and Eli Upfal — *Probability and Computing: Randomization and Probabilistic Techniques in Algorithms and Data Analysis*, 2nd ed., Cambridge University Press, 2017. [Free PDF of 1st edition: https://www.cs.purdue.edu/homes/spa/courses/pg17/mu-book.pdf]

2. J. R. Norris — *Markov Chains*, Cambridge University Press, 1997. [Free: http://www.statslab.cam.ac.uk/~james/Markov/]

3. David Williams — *Probability with Martingales*, Cambridge University Press, 1991.

4. Sheldon Ross — *Introduction to Probability Models*, 10th ed., Academic Press, 2010.

5. Ben Lambert — Bayesian Statistics lecture series, University of Oxford. [Free: https://www.youtube.com/c/SpartacanUsuals]

6. NPTEL — Stochastic Processes, IIT Bombay. [Free: https://nptel.ac.in]